

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	25-06-2025
Team ID	23A51A4250
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The screenshot displays the ServiceNow Workflow Studio interface for the 'Metro Ticket Booking Flow'. The workflow is currently in 'Active' status. The 'TRIGGER' section is set to 'Service Catalog'. The 'ACTIONS' section contains two steps: '1. Get Catalog Variables from Book A Metro Ticket from Book A Metro Ticket' and '2. Create Metro Database Record'. An 'ERROR HANDLER' section is also present. The 'Data' panel on the right lists various variables, including 'Requested Item Record', 'Run Start Time UTC', 'Table Name', 'Run Start DateTime', and several catalog variables like 'enter_payment_mode', 'amount_for_single_journey', 'no_of_passengers', 'amount_including_return', 'journey_type', 'mode_of_payment', 'passenger_name', 'travel_date', 'startings_from', and 'going_to'.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
------	-----------	-------------	------------

1.	User Interface	Interface for passengers to book metro tickets	ServiceNow Service Catalog, Portal UI
2.	Application Logic-1	Fare calculation based on source and destination stations	Catalog Client Scripts (JavaScript)
3.	Application Logic-2	Server-side fare retrieval and validation	Script Includes, GlideAjax
4.	Application Logic-3	Automated ticket booking workflow and notifications	Flow Designer, Business Rules
5.	Database	Stores metro stations, fares, and booking information	ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item)
6.	Cloud Database	Cloud-based data storage managed by ServiceNow	ServiceNow Cloud Database
7.	File Storage	Stores QR codes, ticket attachments, and records	ServiceNow Attachments
8.	External API-1	QR Code generation for digital tickets	QR Code Generator API / Scripted QR Generation
9.	External API-2	Email notification service for ticket confirmation	ServiceNow Notification Engine
10.	Machine Learning Model	Ticket booking trends, revenue, and usage reports	ServiceNow Reports & Dashboards
11.	Infrastructure (Server / Cloud)	Application hosting and execution environment	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Client-side scripting and customization used for fare calculation and form validation	JavaScript, GlideAjax

2.	Security Implementations	Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission	ServiceNow ACLs, User Roles, HTTPS, IAM Controls
3.	Scalable Architecture	ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes	ServiceNow SaaS Architecture, MultiTenant Cloud Platform
4.	Availability	High availability through ServiceNow cloud infrastru managed services	ServiceNow Cloud Infrastructure, cture with continuous monitoring and Redundant Data Centers
5.	Performance	Fast ticket booking process through server-side pro cloud-hosted services	GlideAjax, Script Includes, ServiceNow cessing, optimized database queries, and Database Optimization

References: <https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html> <https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html> https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html <https://developer.servicenow.com> <https://www.servicenow.com/community/>